

Research Statement

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I am a scholar of political communication and behavior who develops methodological innovations to study media and politics. My research demonstrates that social media's effects are far broader in scope than the spread of misinformation or hyperpartisan news. Social media uniquely provides streams of public audience feedback and reactions, allowing readers to form a picture of the preferences and opinions of other citizens in real time. My job market paper combines large-scale observational data with original experiments to show how social media narratives differ from those in mainstream news outlets, and how this contrast causes media organizations and experts to appear out of touch with the public. Beyond social media, I am broadly interested in how features of the modern information landscape such as algorithmic content ranking, evolving journalistic norms, and the rise of artificial intelligence affect political attitudes and beliefs. My methodological research complements this substantive work, as studying contemporary information environments requires new forms of measurement and experimentation. These research strands have produced X published or forthcoming peer reviewed articles during my graduate studies.

Media shapes perceptions of others

I study how media affects politics by shaping the way audiences perceive others. This stands in contrast to traditional theories which focus primarily on persuasion — eg, how news reports or political speeches directly sway audiences' opinions. The key insight driving this research agenda is that citizens use media to make inferences about politically relevant actors including politicians, experts, media organizations, and their fellow citizens. Different features of the contemporary information environment, such as social media, algorithmic

content ranking, journalistic norms, artificial intelligence can both inform and distort these perceptions of others. These perceptions can then have downstream effects on political attitudes and behavior such as trust in institutions, economic perceptions, and polarization.

I develop this argument in my job market paper, *Out of Touch: How Economic News and Commentary Reduce Trust in Media*, focusing on social media’s role in allowing readers to make inferences about public opinion from online discussion of the economy in the United States. I use computational methods to describe the tone of online economic discussion on Reddit, a social media platform and news aggregator and compare it to the tenor of mainstream economic news coverage. The analysis shows that the tone of online commentary declined precipitously starting in 2021 (along with American consumer sentiment), while mainstream news remained steadfastly upbeat. To test the implications of this mismatch, I conduct an experiment which shows people economic news articles accompanied by user comments, as they would encounter on news websites and social media platforms. Changing the tone of user comments substantially shaped participants’ perceptions of both economic conditions and public opinion about those conditions, even more so than changing the tone of the news article. Moreover, the results show that mismatch affects trust in media and experts. Readers who reported being dissatisfied with the economy prior to seeing an upbeat news article became more distrustful of media and experts, while pairing upbeat articles with negative user comments caused declines in trust for all readers.

In *Digital Rubbernecking: Attention-based versus Engagement-based Ranking*, coauthored with Jennifer Allen and Chris Barrie, we then tackle the question of how the ranking algorithms employed by social media platforms distort users’ perceptions of the relative extremism of others. Contemporary social media algorithms are now highly attuned to what audiences pay attention to, in addition to what they engage with. We contend that these attention-driven algorithms surface disproportionately toxic content from both sides of the ideological aisle in a process we term “digital rubbernecking.” While this process may break the echo chamber that concerned previous scholars, we seek to bring attention to a different kind of harm: a distortion of users’ perceptions of the extremism of others. Two methodological innovations allow us to test this argument. First, we collect observational data from real world social media platform use enabled by custom browser software I developed. Second,

we use an experimental design that leverages an implementation of a collaborative-filtering content ranking algorithm, similar to the algorithms used by major social media platforms. This design allows us to directly compare algorithms optimized for attention versus other objectives such as engagement or congruence with explicit user preferences and analyze their effects on the perceived extremism of other citizens.

This research stream is not only limited to social media. In *When Coverage Diverges from Politics: Source Selection in Climate News*, coauthored with Hye Young You, we explore how changes in journalistic norms affects how mainstream news outlets portray the urgency of the climate crisis. This shift potentially causes readers to misjudge the motivations behind opponents of climate policy. Combining large-scale text analysis, large language models, campaign finance data, and congressional speech records, we show that mainstream news coverage places substantially greater emphasis on the urgency of climate action than either mass public opinion or congressional debate would suggest. We argue that this distortion is driven in part by changes in the composition of the journalistic profession itself, where younger and less experienced journalists are less likely to adhere to the traditional balance norm that governed legacy news organizations. Our analysis shows that newer journalists are systematically more ideologically extreme in their sourcing decisions, suggesting that changes in professional selection increasingly shape the informational signals citizens receive about the distribution of political beliefs in society.

In future work, I plan to explore how artificial intelligence and news mediated by large language models shapes or distorts perceptions of public opinion. In contrast to social media, news mediated by artificial intelligence presents information in a flat and source-less manner, potentially reducing the ability of audiences to make inferences about the preferences and opinions of other groups. Its singular omniscient voice compresses diverse viewpoints, obscuring the distribution of beliefs that exist within society. Moreover, while flagship AI systems use retrieval augmented generation to synthesize information from a wide variety of sources little is known about how the choice of sources to consult affects the extent to which users are able to make inferences about those sources.

Digital communication, polarization, and participation

I am also broadly interested in how digital communication systems shape political polarization and participation. A central question in political communication is whether media effects are inherently limited or whether minimal effects reflect the difficulty of studying communication in realistic settings; a particular challenge in studying digital platforms is finding tractable ways of delivering messages at scale to experimental subjects.

Before embarking on my PhD studeies, I was able to work with the Acronym team to study whether large-scale digital communication can influence political participation. In *A 2 Million-Person, Campaign-Wide Field Experiment Shows How Digital Advertising Affects Voter Turnout* (with Jennifer Allen, Minali Aggarwal, Alex Coppock, and others; published in *Nature Human Behavior*), we conducted an \$8.9 million field experiment among two million moderate and low-information voters during the 2020 presidential election. Despite the unprecedented scale and sustained exposure, we found limited average effects on turnout. However, we uncovered encouraging heterogeneity — political ads based on news items appeared to have stronger mobilizing and persuasive effects.

These findings motivate an ongoing project, *Does Out-Partisan Social Media News Exposure Alleviate Polarization?*. However, without massive budgets or platform collaboration, large scale, ecologically valid social media experiments are near impossible to implement. I address this challenge by developing a browser extension to experimentally manipulate exposure to partisan news in users’ real Twitter/X feeds. By changing the content users encounter over an extended period while preserving the structure of their existing online environments, this project provides a strong test of how information exposure affects polarization and political attitudes. In *The Prosocial Ranking Challenge: Reducing Polarization on Social Media without Sacrificing Engagement* (with Jonathan Stray, George Beknazar-Yuzbashev, Kylan Rutherford, Mateusz Stalinski, and others), we use a similar browser extenstion approach to experimentally evaluate alternative ranking algorithms across multiple social media platforms during the 2024 presidential election. We find that alternative ranking systems reduced affective polarization while preserving engagement, demonstrating that the organization of digital communication environments can influence political outcomes.

Computational Methods

Answering these research questions requires methodological innovations as studying contemporary information environments requires new forms of measurement and experimentation. Contemporary political communication poses two challenges: researchers must analyze increasingly unstructured and large-scale textual datasets and also must study behavior embedded within increasingly complex platform environments.

One strand of my work advances text-as-data methods for measuring political concepts in large collections of text. In *Applications of GPT in Political Science Research: Extracting Information from Unstructured Text* (coauthored with Hye Young You and others, published in *PS: Political Science and Politics*), we evaluate the use of large language models for information extraction and text classification in political science research. In *Ambiguity-Robust Dictionaries using Contextualized Embeddings*, coauthored with fellow NYU Politics student Jamil Citavarese, we introduce a method that leverages contextual language embeddings to distinguish relevant from irrelevant uses of dictionary keyword terms. We argue that keyword based approaches are still valuable in contexts where interpretability, reproducibility, and low costs are paramount and show that our approach improves measurement precision while preserving the key advantages of dictionary methods. Moving forward, I plan to continue developing methodological tools that both harness the power of large language models as well as auditing and complementing their weaknesses.

A second strand develops new approaches for measuring and manipulating digital information exposure in ecologically realistic ways. As described in my projects that explore the effect of partisan content and alternative ranking systems on polarization, I developed a set of browser-based experimental tools that allow researchers to measure exposure patterns and manipulate ranked content in users' existing social media feeds. These tools make it possible to audit real-world algorithmic outputs, measure patterns of information exposure, and experimentally manipulate the ranking systems participants encounter online.

Together, this methodological work reflects a broader commitment to developing scalable and transparent tools that expand the discipline's ability to study how emerging technologies shape political behavior.